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CO1 – AT2

MLA0201 – Fundamentals of Machine Learning

Concept Mapping

1) Construct a concept map illustrating the complete Machine Learning workflow for predicting house prices using suitable ML algorithm.

A)

Definition:

House Price Prediction is a supervised machine learning regression problem that predicts the selling price of a house using historical housing data. Since the output is a continuous numerical value (price), regression algorithms such as Linear Regression are commonly used.

Description:

House Price Prediction helps estimate the market value of a house by analysing various features such as location, area, number of bedrooms, age of the house, nearby facilities, and parking availability. The machine learning model learns patterns from previous housing records and predicts the expected price of a new house. This improves pricing accuracy and supports buyers, sellers, and real estate agencies in making better decisions.

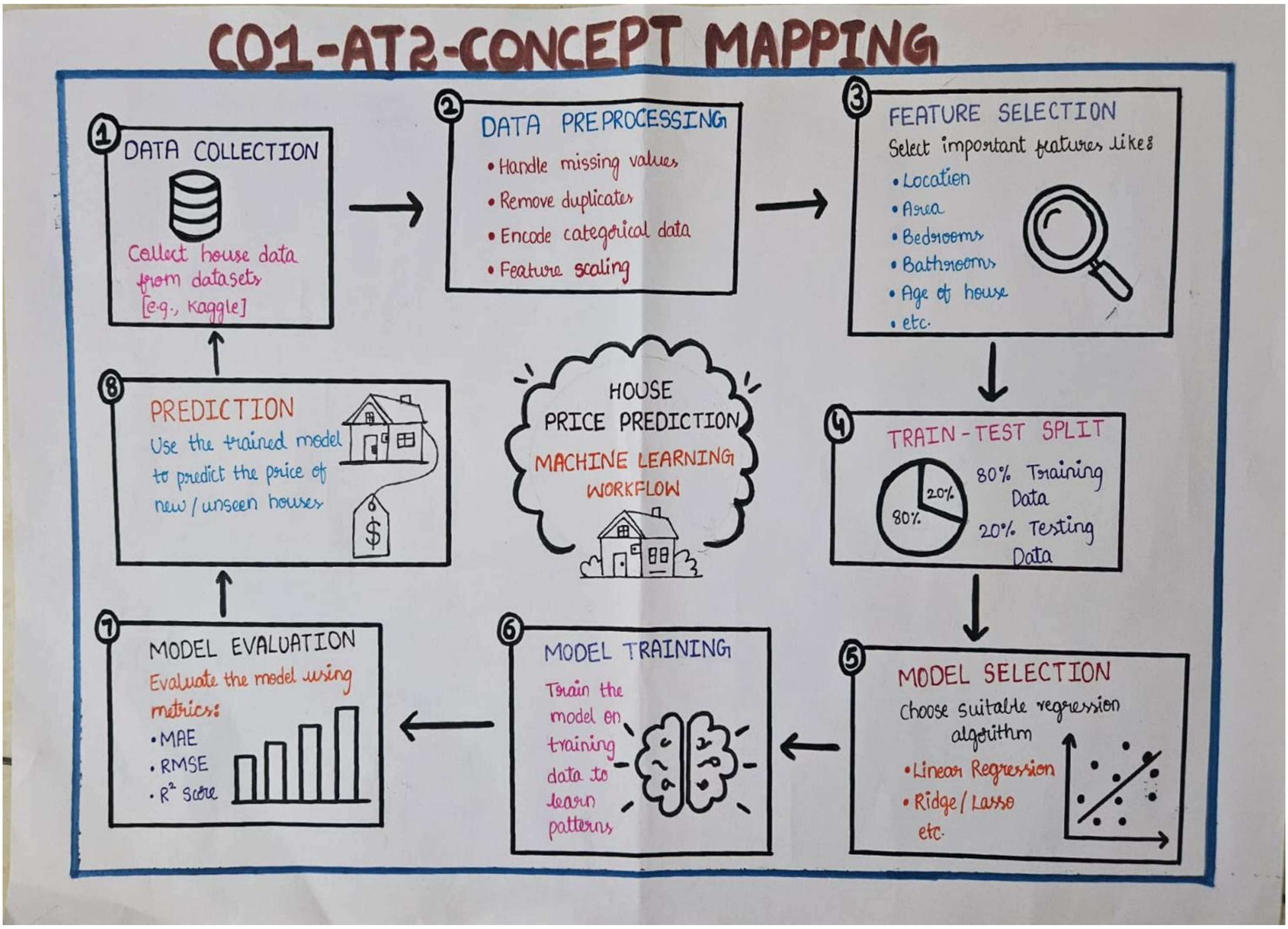
Applications:

- Real estate price estimation
- Property investment analysis
- Online housing websites (99acres, MagicBricks, Zillow)
- Bank loan valuation
- Government property assessment

Suitable Dataset:

- Boston Housing Dataset
- California Housing Dataset
- Kaggle House Price Prediction Dataset

Concept Map:



2) Construct a concept map demonstrating how Classification algorithms are used in Medical Disease Prediction (e.g., Diabetes or Heart Disease Prediction).

A)

Definition:

Medical Disease Prediction is a supervised machine learning classification problem that predicts whether a patient has a disease or not using medical records and health parameters. Classification algorithms classify patients into categories such as Disease Present or Disease Absent.

Description:

Machine Learning helps doctors identify diseases at an early stage by analysing patient information like glucose level, blood pressure, cholesterol, BMI, age, heart rate, and medical history. Classification algorithms learn from previously diagnosed patient data and accurately predict disease status for new patients. This enables faster diagnosis, better treatment planning, and improved healthcare services.

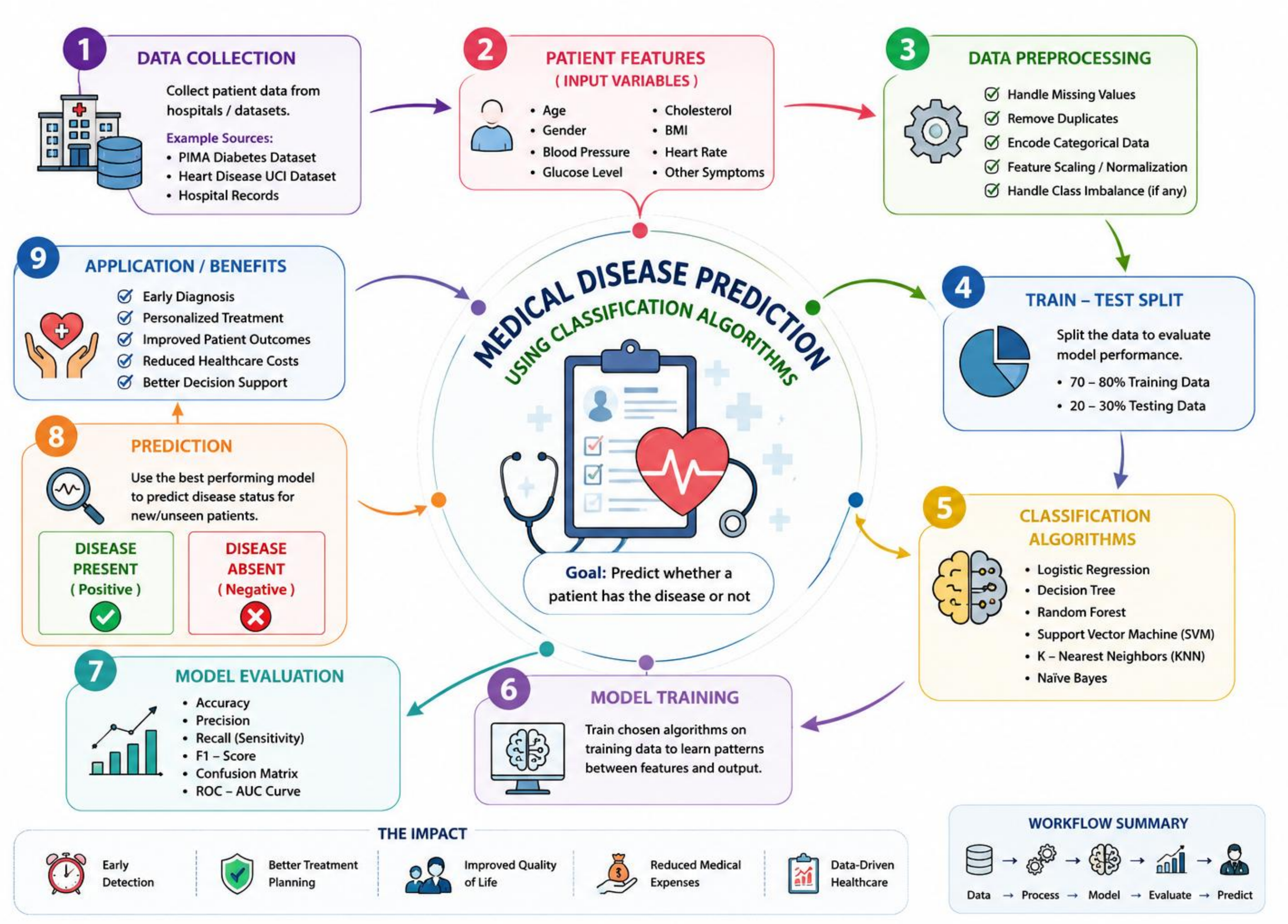
Applications:

- Diabetes Prediction
- Heart Disease Prediction
- Cancer Detection
- Kidney Disease Prediction
- Hospital Decision Support Systems

Suitable Dataset:

- PIMA Indians Diabetes Dataset
- Heart Disease UCI Dataset
- Breast Cancer Wisconsin Dataset

Concept Map:



3) Construct a concept map showing how Machine Learning is applied in Credit Card Fraud Detection.

A)

Definition:

Credit Card Fraud Detection is a binary classification problem where machine learning identifies whether a credit card transaction is genuine or fraudulent by learning transaction patterns from historical data.

Description:

Millions of online transactions occur every day, making manual fraud detection impossible. Machine learning analyses transaction amount, location, merchant details, transaction frequency, and customer behaviour to detect suspicious activities. When unusual patterns are found, the system immediately alerts the bank or blocks the transaction, thereby reducing financial losses.

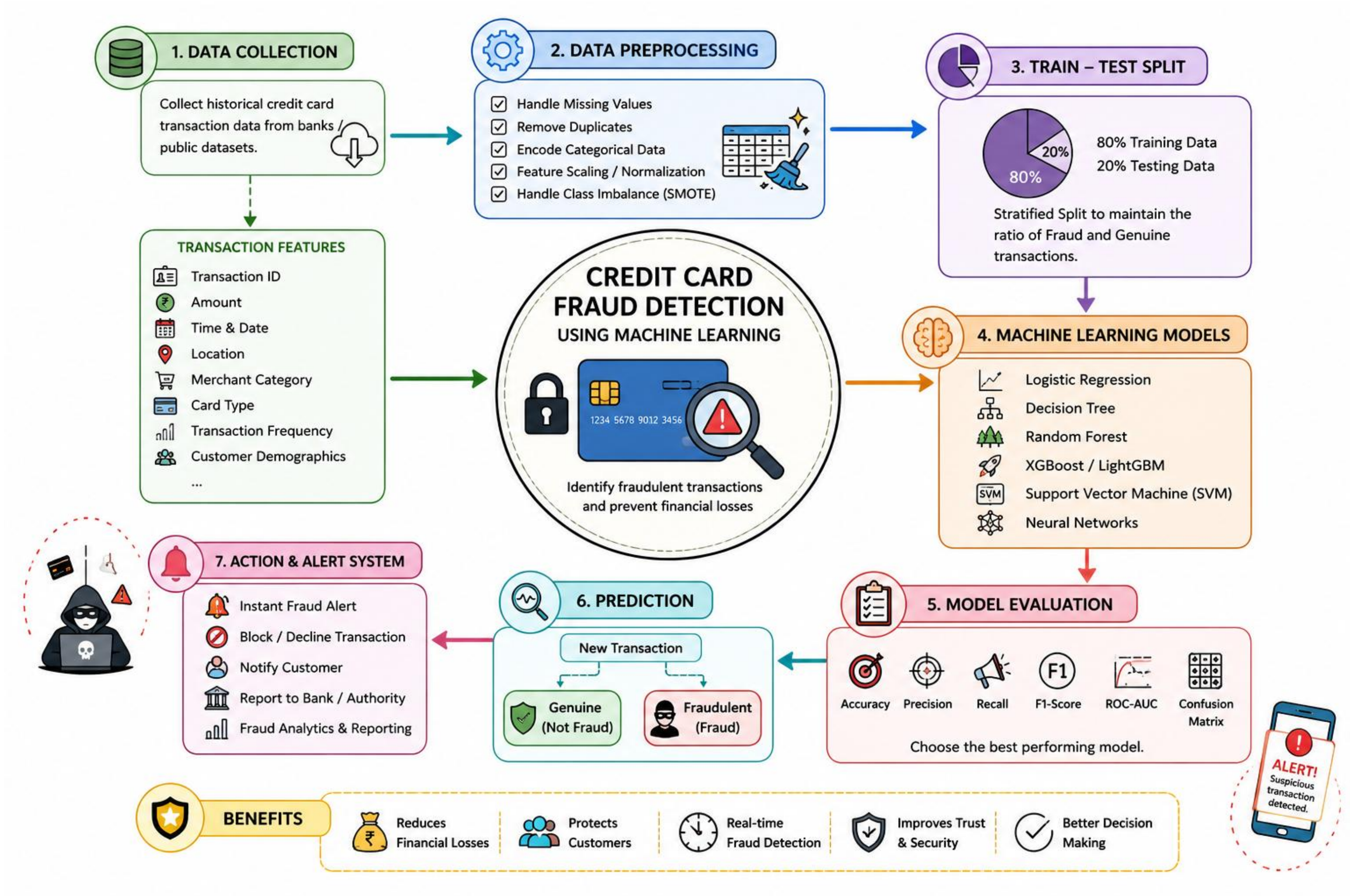
Applications:

- Online Banking Security
- Credit Card Fraud Detection
- Digital Payment Systems
- E-commerce Transactions
- Financial Cybersecurity

Suitable Dataset:

- Kaggle Credit Card Fraud Detection Dataset
- IEEE-CIS Fraud Detection Dataset

Concept Map:



4) Construct a concept map explaining Student Performance Prediction using the Multiple Linear Regression algorithm.

A)

Definition:

Student Performance Prediction is a supervised regression problem where Multiple Linear Regression predicts students' final marks using multiple independent variables such as attendance, study hours, assignment scores, previous marks, and internal assessment marks.

Description:

Educational institutions use machine learning to identify students who may perform well or require additional support. Multiple Linear Regression studies the relationship between several input features and the final examination score. The predicted results help teachers monitor academic progress, provide personalized guidance, and improve overall student performance.

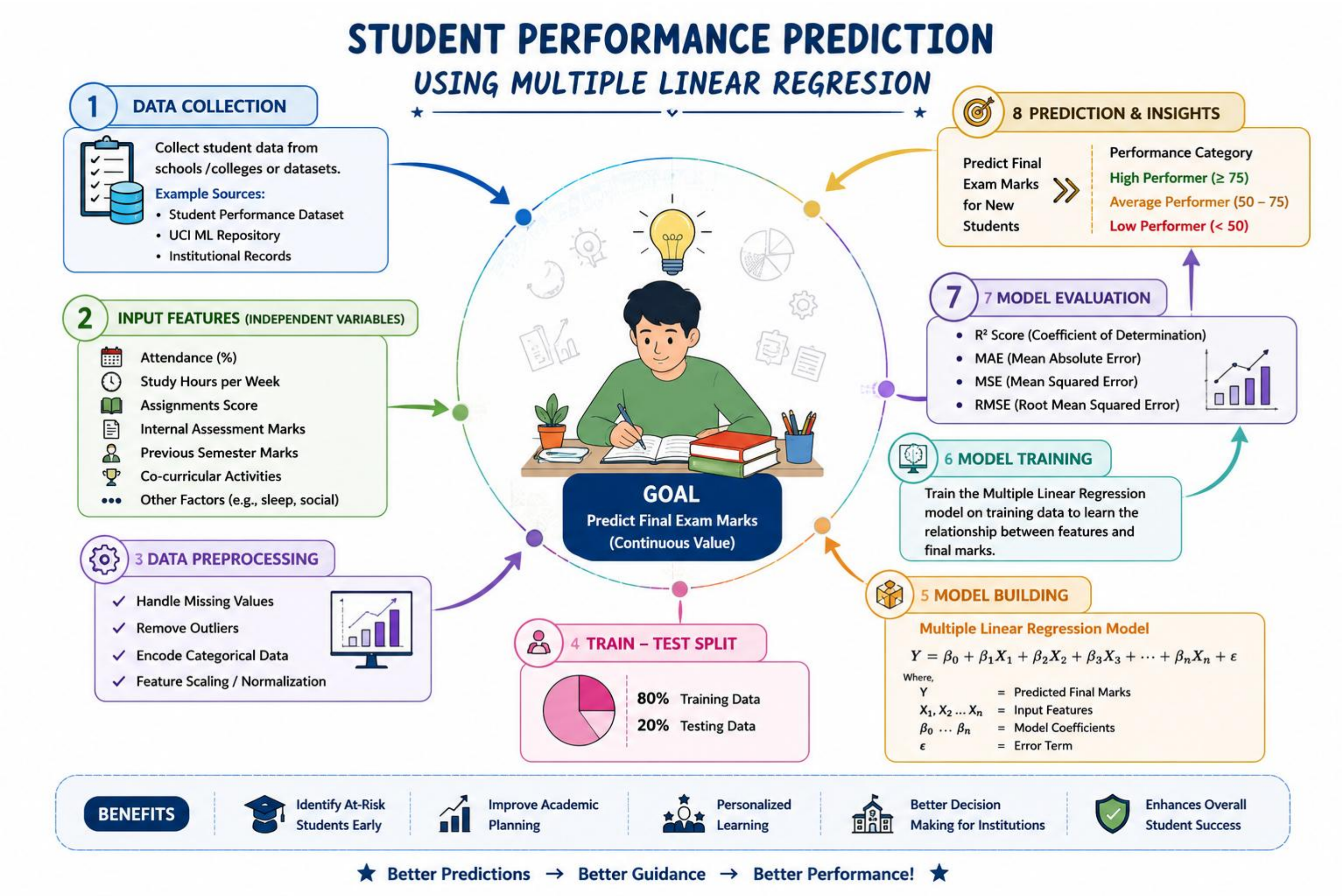
Applications:

- Academic Performance Analysis
- Early Identification of Weak Students
- Personalized Learning Systems
- Educational Analytics
- Student Counselling

Suitable Dataset:

- Student Performance Dataset (UCI)
- Kaggle Student Performance Dataset

Concept Map:



5) Construct a concept map showing how Decision Tree Classification is used for Loan Approval Prediction.

A)

Definition:

Loan Approval Prediction is a supervised classification problem that predicts whether a loan application should be approved or rejected using the Decision Tree Classification algorithm based on applicant information.

Description:

Banks receive thousands of loan applications every day. Machine learning helps evaluate applicants by analysing income, credit history, employment status, loan amount, repayment history, and existing debts. The Decision Tree algorithm creates decision rules that classify applications into approved or rejected categories, improving speed, consistency, and fairness in loan processing.

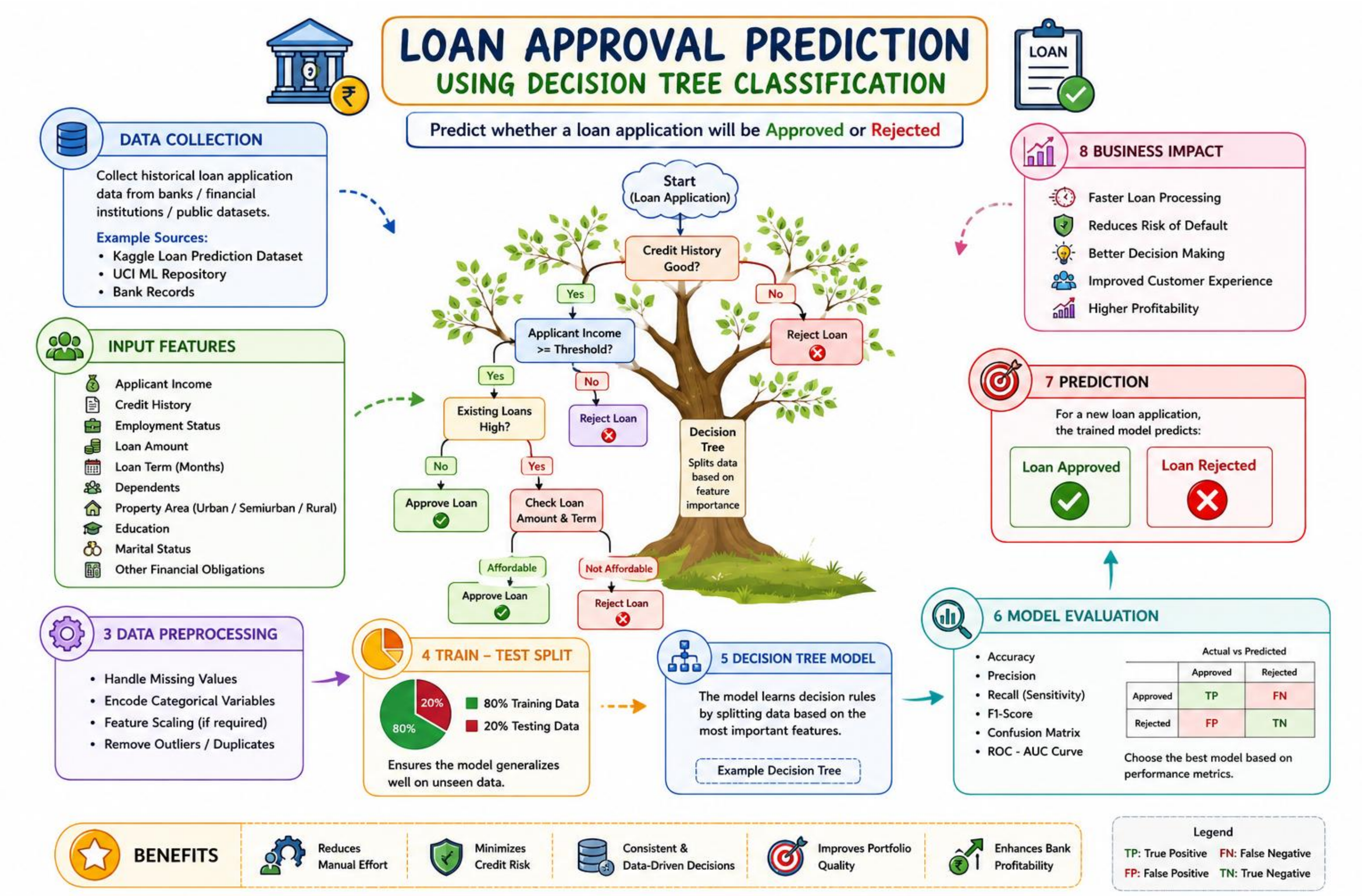
Applications:

- Bank Loan Approval
- Credit Risk Analysis
- Financial Decision Support
- Insurance Risk Assessment
- Customer Credit Evaluation

Suitable Dataset:

- Kaggle Loan Prediction Dataset
- UCI Credit Approval Dataset

Concept Map:



LOAN APPROVAL PREDICTION
USING DECISION TREE CLASSIFICATION
Predict whether a loan application will be Approved or Rejected
LOAN
DATA COLLECTION
Collect historical loan application data from banks / financial institutions / public datasets.
Example Sources:
• Kaggle Loan Prediction Dataset
• UCI ML Repository
• Bank Records
INPUT FEATURES
Applicant Income
Credit History
Employment Status
Loan Amount
Loan Term (Months)
Dependents
Property Area (Urban / Semiurban / Rural)
Education
Marital Status
Other Financial Obligations
Start (Loan Application)
Credit History Good?
Yes
No
Reject Loan
Applicant Income >= Threshold?
Yes
No
Reject Loan
Existing Loans High?
No
Yes
Approve Loan
Check Loan Amount & Term
Affordable
Not Affordable
Approve Loan
Reject Loan
Decision Tree
Splits data based on feature importance
8 BUSINESS IMPACT
Faster Loan Processing
Reduces Risk of Default
Better Decision Making
Improved Customer Experience
Higher Profitability
7 PREDICTION
For a new loan application, the trained model predicts:
Loan Approved
Loan Rejected
3 DATA PREPROCESSING
• Handle Missing Values
• Encode Categorical Variables
• Feature Scaling (if required)
• Remove Outliers / Duplicates
4 TRAIN – TEST SPLIT
20%
80%
80% Training Data
20% Testing Data
Ensures the model generalizes well on unseen data.
5 DECISION TREE MODEL
The model learns decision rules by splitting data based on the most important features.
Example Decision Tree
6 MODEL EVALUATION
• Accuracy
• Precision
• Recall (Sensitivity)
• F1-Score
• Confusion Matrix
• ROC - AUC Curve
Actual vs Predicted
Approved
Rejected
Approved TP FN
Rejected FP TN
Choose the best model based on performance metrics.
BENEFITS
Reduces Manual Effort
Minimizes Credit Risk
Consistent & Data-Driven Decisions
Improves Portfolio Quality
Enhances Bank Profitability
Legend
TP: True Positive
FN: False Negative
FP: False Positive
TN: True Negative